CHAPTER 4 [NO 1-32]

DC BIASING - BJT

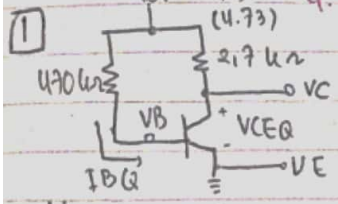
ELEKTRONIKA ANALOG

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JAWABAN SAJA

4.3 Fixed-Bias

1 Fig (4.73)



$\beta = 90$

a) $I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16\,V - 0{,}7\,V}{470\,k\Omega} = \frac{15{,}3\,V}{470\,k\Omega} = 32{,}55\,\mu A$

b) $I_{CQ} = \beta I_{BQ} = 90 \times 32{,}55\,\mu A = 2{,}93\,mA$

c) $V_{CEQ} = V_{CC} - I_{CQ} R_C = 16\,V - (2{,}93\,mA)(2{,}7\,k\Omega)$
$= 8{,}09\,V$

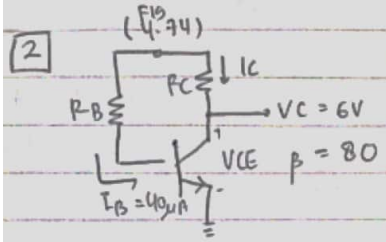
d) $V_C = V_{CEQ} = 8{,}09\,V$

e) $V_B = V_{BE} = 0{,}7\,V$

f) $V_E = 0\,V$

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2 Fig (4.74)



$\beta = 80$

Tentukan

a) $I_C = \beta I_B = 80 \times 40\,\mu A = 3{,}2\,mA$

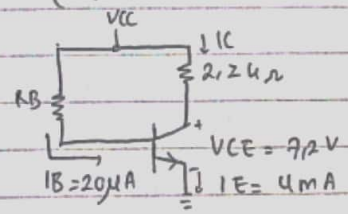
b) $R_C = \frac{V_{RC}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{12\,V - 6\,V}{3{,}2\,mA} = 1{,}875\,k\Omega$

c) $R_B = \frac{V_{RB}}{I_B} = \frac{12\,V - 0{,}7\,V}{40\,\mu A} = 282{,}5\,k\Omega$

d) $V_{CE} = V_C = 6V$

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3 (Fig 4.75)



(a) $I_C = I_E - I_B = 4\,mA - 20\,\mu A = 3{,}98\,mA \cong 4\,mA$

(b) $V_{CC} = V_{CE} + I_C R_C = 7{,}2\,V + (3{,}98\,mA)(2{,}2\,k\Omega)$
$= 15{,}96\,V \cong 16\,V$

(c) $\beta = \frac{I_C}{I_B} = \frac{3{,}98\,mA}{20\,\mu A} = 199 \cong 200$

(d) $R_B = \frac{V_{RB}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{15{,}96\,V - 0{,}7\,V}{20\,\mu A} = 763\,k\Omega$

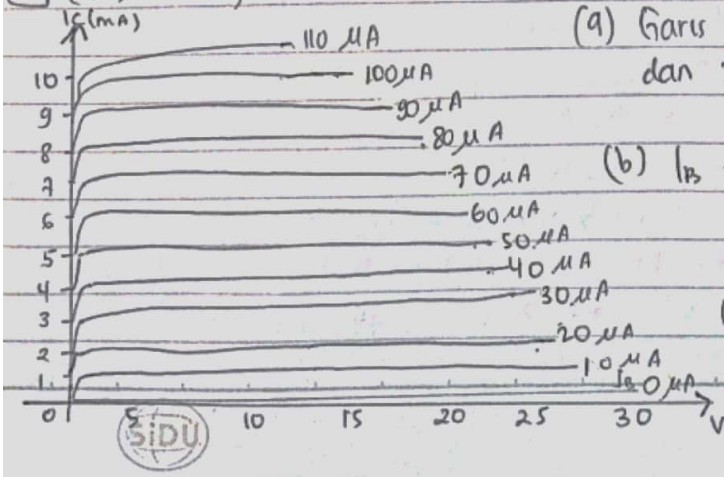
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4 Menentukan I_C sat fig 4.73

$I_{C\,sat} = \frac{V_{CC}}{R_C} = \frac{16\,V}{2{,}7\,k\Omega} = 5{,}93\,mA$

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5 (Fig. 4.76)



(a) Garis beban memotong sumbu vertikal $IC = \frac{21\,V}{3\,k\Omega} = 7\,mA$ dan sumbu horizontal $VCE = 21\,V$

(b) $I_B = 25\,\mu A$: $R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{21\,V - 0{,}7\,V}{25\,\mu A} = 812\,k\Omega$

(c) $I_{CQ} \cong 3{,}4\,mA$; $VCE \cong 10{,}95\,V$

(d) $\beta = \frac{I_C}{I_B} = \frac{3{,}4\,mA}{25\,\mu A} = 136$

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(e) $\alpha = \frac{\beta}{\beta+1} = \frac{136}{136+1} = \frac{136}{137} = 0{,}992$

(f) $I_{C_{sat}} = \frac{V_{CC}}{R_C} = \frac{21\,V}{3\,k\Omega} = 7\,mA$

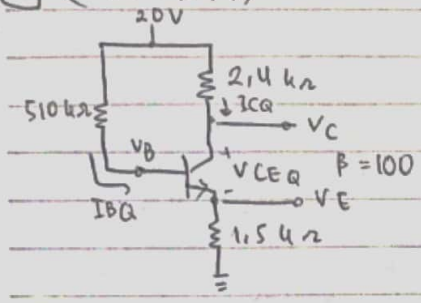
(h) $P_D = V_{CEQ} \cdot I_{CQ} = (10{,}75\,V)(3{,}4\,mA) = 36{,}55\,mW$

(i) $P_S = V_{CC}(I_C + I_B) = 21\,V\,(3{,}4\,mA + 25\,\mu A) = 71{,}92\,mW$

(j) $P_F = P_S - P_D = 71{,}92\,mW - 36{,}55\,mW = 35{,}37\,mW$

4.4 Emitter Stabilized Bias Circuit

6 (Fig. 4.77)



a) $I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{20\,V - 0{,}7\,V}{510\,k\Omega + (101)1{,}5\,k\Omega} = 29{,}18\,\mu A$

b) $I_{CQ} = \beta I_{BQ} = 100 \times 29{,}18\,\mu A = 2{,}92\,mA$

c) $V_{CEQ} = V_{CC} - I_C(R_C + R_E) = 20\,V - (2{,}92\,mA)(2{,}4\,k\Omega + 1{,}5\,k\Omega)$

$= 8{,}61\,V$

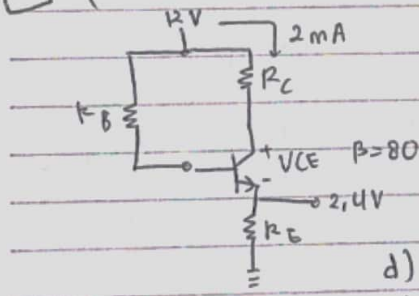
d) $V_C = V_{CC} - I_C R_C = 20\,V - (2{,}92\,mA)(2{,}4\,k\Omega)$

$= 13\,V$

e) $V_B = V_{CC} - I_B R_B = 20\,V - (29{,}18\,\mu A)(510\,k\Omega) = 5{,}12\,V$

f) $V_E = V_C - V_{CE} = 13\,V - 8{,}61\,V = 4{,}39\,V$

7 (Fig. 4.78)



(a) $R_C = \frac{V_{CC} - V_C}{I_C} = \frac{12\,V - 7{,}6\,V}{2\,mA} = \frac{4{,}4\,V}{2\,mA} = 2{,}2\,k\Omega$

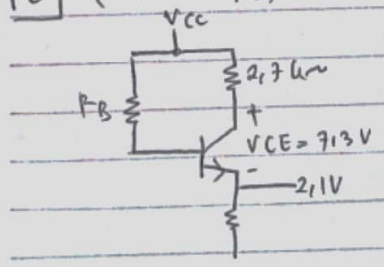
b) $I_E \approx I_C$: $R_E = \frac{V_E}{I_E} = \frac{2{,}4\,V}{2\,mA} = 1{,}2\,k\Omega$

c) $R_B = \frac{V_{RB}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{12\,V - 0{,}7\,V - 2{,}4\,V}{2\,mA/80} = 356\,k\Omega$

d) $V_{CE} = V_C - V_E = 7{,}6\,V - 2{,}4\,V = 5{,}2\,V$

e) $V_B = V_{BE} + V_E = 0{,}7\,V + 2{,}4\,V = 3{,}1\,V$

8 (Fig. 4.79)



(a) $I_C \approx I_E = \frac{V_E}{R_E} = \frac{2{,}1\,V}{0{,}68\,k\Omega} = 3{,}09\,mA$

$\beta = \frac{I_C}{I_B} = \frac{3{,}09\,mA}{20\,\mu A} = 154{,}5$

(b) $V_{CC} = V_{RC} + V_{CE} + V_E$

$= 3{,}09\,mA \times 2{,}7\,k\Omega + 7{,}3\,V + 2{,}1\,V$

$= 17{,}74\,V$

(c) $R_B = \frac{V_{RB}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{17{,}74\,V - 0{,}7\,V - 2{,}1\,V}{20\,\mu A} = 747\,k\Omega$

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9 (Fig. 4.77)

$$I_{Csat} = \frac{V_{CC}}{R_C + R_E} = \frac{20\,V}{2{,}4\,k\Omega + 1{,}5\,k\Omega} = 5{,}13\,mA$$

10 (Fig. 4.76) Dik. $I_{CQ} = 4\,mA$ dan $V_{CEQ} = 10\,V$

a) $$I_{Csat} = 6{,}8\,mA = \frac{V_{CC}}{R_C + R_E} = \frac{24\,V}{R_C + 1{,}2\,k\Omega}$$

$$R_C + 1{,}2\,k\Omega = \frac{24\,V}{6{,}8\,mA} = 3{,}529\,k\Omega$$

$$R_C = 2{,}33\,k\Omega$$

b) $$\beta = \frac{I_C}{I_B} = \frac{4\,mA}{30\,\mu A} = 133{,}33$$

c) $$R_B = \frac{V_{RB}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{24\,V - 0{,}9\,V - (4\,mA)(1{,}2\,k\Omega)}{30\,\mu A} = 616{,}67\,k\Omega$$

d) $$P_D = V_{CEQ} \cdot I_{CQ}$$
$$= 10\,V \times 4\,mA = 40\,mW$$

e) $$P = I_C^2 \cdot R_C = (4\,mA)^2 \times (2{,}33\,k\Omega) = 37{,}28\,mW$$

11 (Fig. 4.73) untuk soal (a – c) dan (Fig. 4.77) untuk soal (d – g)

(a) $I_{CQ} = 2{,}93\,mA$, $V_{CEQ} = 8{,}09\,V$

(b) $I_{BQ} = 32{,}55\,\mu A$ (rama)

$$I_{CQ} = \beta I_{BQ} = (135)(32{,}55\,\mu A) = 4{,}39\,mA$$

$$V_{CEQ} = V_{CC} - I_{CQ} \cdot R_C = 16\,V - (4{,}39\,mA)(2{,}7\,k\Omega) = 4{,}15\,V$$

(c) $$\%\,\Delta I_C = \left|\frac{4{,}39\,mA - 2{,}93\,mA}{2{,}93\,mA}\right| \times 100\,\% = 49{,}83\,\%$$

$$\%\,\Delta V_{CE} = \left|\frac{4{,}15\,V - 8{,}09\,V}{8{,}09\,V}\right| \times 100\,\% = 48{,}70\,\%$$

Kurang dari 50 % karena tingkat akurasi yg dilakukan melalui perhitungan

(d) $I_{CQ} = 2{,}02\,mA$, $V_{CEQ} = 8{,}61\,V$ ($I_{BQ} = 20{,}18\,\mu A$)

(e) $$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{20\,V - 0{,}9\,V}{510\,k\Omega + (150+1)(1{,}5\,k\Omega} = 26{,}21\,\mu A$$

$$I_{CQ} = \beta I_{BQ} = 150 \times 26{,}21\,\mu A = 3{,}93\,mA$$

$$V_{CEQ} = V_{CC} - I_C (R_C + R_E)$$
$$= 20\,V - (3{,}93\,mA)(2{,}4\,k\Omega + 1{,}5\,k\Omega) = 4{,}67\,V$$

(f) $$\%\,\Delta I_C = \left|\frac{3{,}93\,mA - 2{,}92\,mA}{2{,}92\,mA}\right| \times 100\,\% = 34{,}59\,\%$$

$$\%\,\Delta V_{CE} = \left|\frac{4{,}67\,V - 8{,}61\,V}{8{,}61\,V}\right| \times 100\,\% = 46{,}76\,\%$$

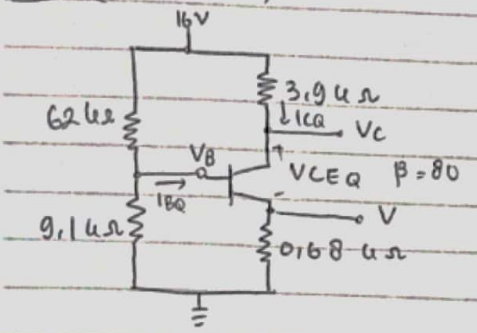
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(g) Untuk IC dan V_{CE} % perubahannya lebih kecil untuk emitor-stabil

4.5 Voltage – Divider Bias

12 (Fig 4.80)



$\beta R_E \overset{?}{\geq} 10 R_2$

$80 \times 0{,}68\ k\Omega \geq 10\,(9{,}1\ k\Omega)$

$54{,}4\ k\Omega \not\geq 91\ k\Omega$ (Tidak)

(a) Menggunakan pendekatan eksak

$$R_{TH} = R_1 // R_2 = 62\ k\Omega // 9{,}1\ k\Omega = 7{,}94\ k\Omega$$

$$E_{TH} = \frac{R_2 V_{CC}}{R_2 + R_1} = \frac{(9{,}1\ k\Omega)(16\ V)}{9{,}1\ k\Omega + 62\ k\Omega} = 2{,}05\ V$$

$$I_{BQ} = \frac{E_{TH} - V_{BE}}{R_{TH} + (\beta+1)R_E} = \frac{2{,}05\ V - 0{,}7\ V}{7{,}94\ k\Omega + 81 \times 0{,}68\ k\Omega} = 21{,}42\ \mu A$$

(b) $I_{CQ} = \beta I_{BQ} = 80 \times 21{,}42\ \mu A = 1{,}71\ mA$

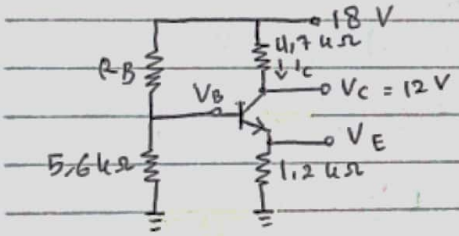
(c) $V_{CEQ} = V_{CC} - I_{CQ}(R_C + R_E) = 16\ V - (1{,}71\ mA)(3{,}9\ k\Omega + 0{,}68\ k\Omega) = 8{,}17\ V$

(d) $V_C = V_{CC} - I_C R_C = 16\ V - (1{,}71\ mA)(3{,}9\ k\Omega) = 9{,}33\ V$

(e) $V_E = I_E R_E \cong I_C R_E = 1{,}71\ mA \times 0{,}68\ k\Omega = 1{,}16\ V$

(f) $V_B = V_E + V_{BE} = 1{,}16\ V + 0{,}7\ V = 1{,}86\ V$

13 (Fig. 4.81)



a) $I_C = \frac{V_{CC} - V_C}{R_C} = \frac{18\ V - 12\ V}{4{,}7\ k\Omega} = 1{,}28\ mA$

b) $V_E = I_E R_E \cong I_C R_E = 1{,}28\ mA \times 1{,}2\ k\Omega = 1{,}54\ V$

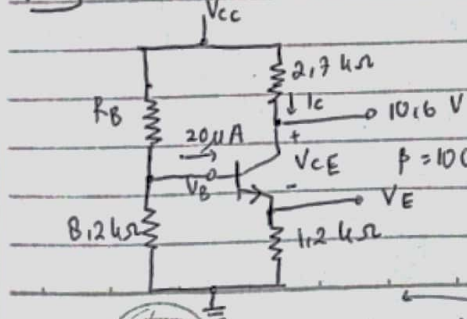
c) $V_B = V_{BE} + V_E = 0{,}7\ V + 1{,}54\ V = 2{,}24\ V$

d) $R_1 = \frac{V_{R1}}{I_{R1}}$; $V_{R1} = V_{CC} - V_B = 18\ V - 2{,}24\ V = 15{,}76\ V$

$$I_{R1} \cong I_{R2} = \frac{V_B}{R_2} = \frac{2{,}24\ V}{5{,}6\ k\Omega} = 0{,}4\ mA$$

$$R_1 = \frac{V_{R1}}{I_R} = \frac{15{,}76\ V}{0{,}4\ mA} = 39{,}4\ k\Omega$$

14 (Fig 4.82)



a) $I_C = \beta I_B = 100 \times 20\ \mu A = 2\ mA$

b) $I_E = I_C + I_B = 2\ mA + 20\ \mu A = 2{,}02\ mA$

$V_E = I_E R_E = 2{,}02\ mA \times 1{,}2\ k\Omega = 2{,}42\ V$

c) $V_{CC} = V_C + I_C R_C = 10{,}6\ V + 2\ mA \times 2{,}7\ k\Omega = 16\ V$

d) $V_{CE} = V_C - V_E = 10{,}6\ V - 2{,}42\ V = 8{,}18\ V$

e) $V_B = V_E + V_{BE} = 2{,}42\ V + 0{,}7\ V = 3{,}12\ V$

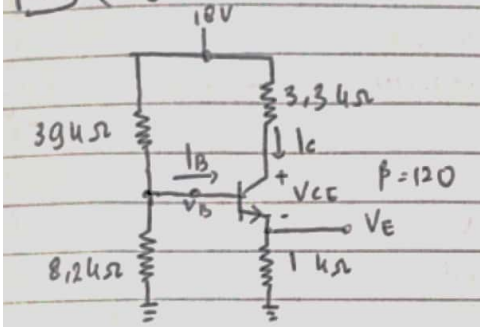
f) $I_{R1} = I_{R2} + I_B = \frac{3{,}12\ V}{8{,}2\ k\Omega} + 20\ \mu A = 400{,}5\ \mu A$

$$R_1 = \frac{V_{CC} - V_B}{I_{R1}} = \frac{16\ V - 3{,}12\ V}{400{,}5\ \mu A} = 32{,}16\ k\Omega$$

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15 $I_{C\,sat} = \dfrac{V_{CC}}{R_C + R_E} = \dfrac{16\,V}{3{,}9\,k\Omega + 0{,}68\,k\Omega} = 3{,}49\,mA$ (Fig. 4.80)

16 (Fig 4.83)



a) $\beta R_E \geq 10 R_2$

$(120)(1\,k\Omega) \geq 10\,(8{,}2\,k\Omega)$

$120\,k\Omega \geq 82\,k\Omega$

- $V_B = \dfrac{R_2 V_{CC}}{R_1 + R_2} = \dfrac{(8{,}2\,k\Omega)(18\,V)}{39\,k\Omega + 8{,}2\,k\Omega} = 3{,}13\,V$

$V_E = V_B - V_{BE} = 3{,}13\,V - 0{,}7\,V = 2{,}43\,V$

$I_C \cong I_E = \dfrac{V_E}{R_E} = \dfrac{2{,}43\,V}{1\,k\Omega} = 2{,}43\,mA$

b) $V_{CE} = V_{CC} - I_C (R_C + R_E)$

$= 18\,V - (2{,}43\,mA)(3{,}3\,k\Omega + 1\,k\Omega) = 7{,}55\,V$

c) $I_B = \dfrac{I_C}{\beta} = \dfrac{2{,}43\,mA}{120} = 20{,}25\,\mu A$

d) $V_E = I_E R_E \cong I_C R_E = 2{,}43\,mA \times 1\,k\Omega = 2{,}43\,V$

e) $V_B = 3{,}13\,V$

17 a) $R_{TH} = R_1 // R_2 = 39\,k\Omega // 8{,}2\,k\Omega = 6{,}78\,k\Omega$

$E_{TH} = \dfrac{R_C V_{CC}}{R_1 + R_2} = \dfrac{8{,}2\,k\Omega\,(18\,V)}{39\,k\Omega + 8{,}2\,k\Omega} = 3{,}13\,V$

$I_B = \dfrac{E_{TH} - V_{BE}}{R_{TH} + (\beta+1) R_E} = \dfrac{3{,}13\,V - 0{,}7\,V}{6{,}78\,k\Omega + (121)(1\,k\Omega)} = 19{,}02\,\mu A$

$I_C = \beta I_B = (120 \times 19{,}02\,\mu A) = 2{,}28\,mA$

b) $V_{CE} = V_{CC} - I_C (R_C + R_E) = 18\,V - (2{,}28\,mA)(3{,}3\,k\Omega + 1\,k\Omega) = 8{,}2\,V$

c) $19{,}02\,\mu A$

d) $V_E = I_E R_E \cong I_C R_E = 2{,}28\,mA \times 1\,k\Omega = 2{,}28\,V$

e) $V_B = V_{BE} + V_E = 0{,}7\,V + 2{,}28\,V = 2{,}98\,V$

Hasilnya menunjukkan bahwa pendekatan perkiraan valid is satisfied

18 (Fig 4.80)

a) $V_B = \dfrac{R_2}{R_1 + R_2} V_{CC} = \dfrac{9{,}1\,k\Omega\,(16V)}{62\,k\Omega + 9{,}1\,k\Omega} = 2{,}05\,V$

$V_E = V_B - V_{BE} = 2{,}05\,V - 0{,}7\,V = 1{,}35\,V$

$I_E = \dfrac{V_E}{R_E} = \dfrac{1{,}35\,V}{0{,}68\,k\Omega} = 1{,}99\,mA$

$I_{CQ} \cong I_E = 1{,}99\,mA$

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$V_{CEQ} = V_{CC} - I_C (R_C + R_E) = 16\,V - (1{,}99\,mA)(3{,}9\,k\Omega + 0{,}68\,k\Omega) = 6{,}89\,V$

$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{1{,}99\,mA}{80} = 24{,}88\,\mu A$

b) $I_{CQ} = 1{,}71\,mA$, $V_{CEQ} = 8{,}17\,V$, $I_{BQ} = 21{,}42\,\mu A$

c) Perbedaan sekitar 14% menunjukkan bahwa pendekatan yg tepat harus digunakan untuk resistansi

19 Diketahui $I_{CQ} = 5\,mA$, $V_{CEQ} = 8\,V$, $V_{CC} = 24\,V$, $R_C = 3R_E$

a) $I_{C\,sat} = 7{,}5\,mA = \frac{V_{CC}}{R_C + R_E} = \frac{24\,V}{3R_E + R_E} = \frac{24\,V}{4R_E}$

$R_E = \frac{24\,V}{4(7{,}5\,mA)} = \frac{24\,V}{30\,mA} = 0{,}8\,k\Omega$

$R_C = 3R_E = 3(0{,}8\,k\Omega) = 2{,}4\,k\Omega$

b) $V_E = I_E R_E \cong I_C R_E = 5\,mA \times 0{,}8\,k\Omega = 4\,V$

c) $V_B = V_E + V_{BE} = 4\,V + 0{,}7\,V = 4{,}7\,V$

d) $V_B = \frac{R_2 V_{CC}}{R_2 + R_1}$, $4{,}7\,V = \frac{R_2(24\,V)}{R_2 + 24\,k\Omega}$ $(\Rightarrow R_2 = 5{,}84\,k\Omega$

e) $\beta_{dc} = \frac{I_C}{I_B} = \frac{5\,mA}{38{,}5\,\mu A} = 129{,}8$

f) $\beta R_E \geq 10 R_2$

$129{,}8 \times 0{,}8\,k\Omega \geq 10(5{,}84\,k\Omega)$

$103{,}84\,k\Omega \geq 58{,}4\,k\Omega$

20 (Fig 4.80)

a) $I_C = 1{,}71\,mA$, $V_{CE} = 8{,}17\,V$

b) Ubah β ke 120 (peningkatan 50%)

$E_{TH} = 2{,}05\,V$, $R_{TH} = 7{,}94\,k\Omega$

$I_B = \frac{E_{TH} - V_{BE}}{R_{TH} + (\beta+1)R_E} = \frac{2{,}05\,V - 0{,}7\,V}{7{,}94\,k\Omega + 121 \times 0{,}68\,k\Omega} = 14{,}96\,\mu A$

$I_C = \beta I_B = 120 \times 14{,}96\,\mu A = 1{,}8\,mA$

$V_{CE} = V_{CC} - I_C(R_C + R_E) = 16\,V - (1{,}8\,mA)(3{,}9\,k\Omega + 0{,}68\,k\Omega) = 7{,}76\,V$

c) $\%\Delta I_C = \left|\frac{1{,}8\,mA - 1{,}71\,mA}{1{,}71\,mA}\right| \times 100\% = 5{,}26\,\%$

$\%\Delta V_{CE} = \left|\frac{7{,}76\,V - 8{,}17\,V}{8{,}17\,V}\right| \times 100\% = 5{,}02\,\%$

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d)

	β_c	β_f	$2\beta_c$
$\%\Delta I_C$	49,83%	34,59%	5,26%
$\%\Delta V_{CE}$	48,70%	46,76%	5,02%
	fixed-bias	Emiter feedback	voltage divider

e) Konfigurasi the voltage-divider adalah yang paling tidak sensitif terhadap perubahan β

21 (fig. 4.83)

1 (a) Pendekatan kira-kira $I_{CQ} = 2,43\ mA$, $V_{CEQ} = 7,55\ V$

Analisis akurat $I_{CQ} = 2,28\ mA$, $V_{CEQ} = 8,2\ V$

Solusi yang akurat akan digunakan untuk mendemonstrasikan pengaruh perubahan.

Menggunakan pendekatan perkiraan menghasilkan $\%\Delta I_C = 0\%$ dan $\%\Delta V_{CE} = 0\%$

(b) $E_{TH} = 3,13\ V$, $R_{TH} = 6,178\ k\Omega$

$$I_B = \frac{E_{TH} - V_{BE}}{R_{TH} + (\beta+1)R_E} = \frac{3,13\ V - 0,7\ V}{6,178\ k\Omega + (180+1)1k\Omega} = 12,94\ \mu A$$

$$I_C = \beta I_B = 180 \times 12,94\ \mu A = 2,33\ mA$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) = 18\ V - (2,33\ mA)(3,3\ k\Omega + 1k\Omega) = 7,98\ V$$

(c) $$\%\Delta I_C = \left|\frac{2,33\ mA - 2,28\ mA}{2,28\ mA}\right| \times 100\% = 2,19\%$$

$$\%\Delta V_{CE} = \left|\frac{7,98\ V - 8,2\ V}{8,2\ V}\right| \times 100\% = 2,68\%$$

Untuk situasi dimana $\beta R_E > 10R_2$ perubahan I_C dan/atau V_{CE} karenda perubahan signifikan dalam akan relatif kecil

(d) $\%\Delta I_C = 2,19\%$ vs 49,83%

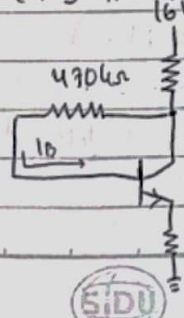
$\%\Delta V_{CE} = 2,68\%$ vs 48,70%

(e) Konfigurasi voltage-divider kurang sensitif

2 $\%\Delta I_C$ dan $\%\Delta V_{CE}$ yang dihasilkan sangat kecil

4.6 Dc Bias with Voltage Feedback

22 (Fig 4.84)



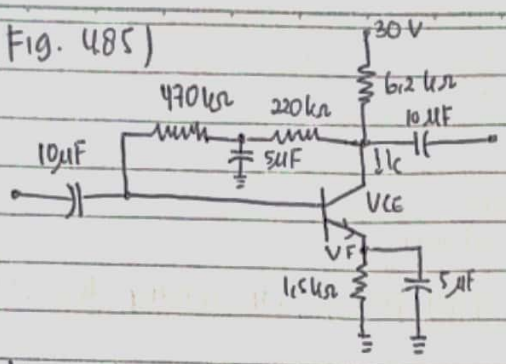
(a) $$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{16\ V - 0,7\ V}{470\ k\Omega + (120)(3,6\ k\Omega + 0,51\ k\Omega)} = 15,88\ \mu A$$

(b) $I_C = \beta I_B = 120 \times 15,88\ \mu A = 1,91\ mA$

(c) $V_C = V_{CC} - I_C R_C = 16\ V - (1,91\ mA)(3,6\ k\Omega) = 9,12\ V$

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23 (Fig. 485)



(a) $I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{30\,V - 0,7\,V}{6,90\,k\Omega + 100(6,2\,k\Omega + 1,5\,k\Omega)}$

$= 20,07\,\mu A$

$I_C = \beta I_B = 100 \times 20,07\,\mu A = 2,01\,mA$

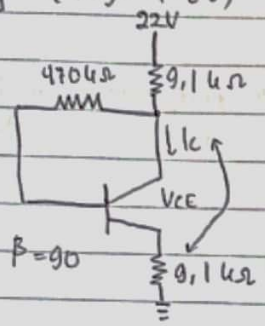
(b) $V_C = V_{CC} - I_C R_C$

$= 30\,V - (2,01\,mA)(6,2\,k\Omega) = 17,54\,V$

(c) $V_E = I_E R_E = 2,01\,mA \times 1,5\,k\Omega = 3,02\,V$

(d) $V_{CE} = V_{CC} - I_C(R_C + R_E) = 30\,V - (2,01\,mA)(6,2\,k\Omega + 1,5\,k\Omega) = 14,52\,V$

24 (Fig. 4.86)



(a) $I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{22\,V - 0,7\,V}{470\,k\Omega + (90)(9,1\,k\Omega + 9,1\,k\Omega)} = 10,09\,\mu A$

$I_C = \beta I_B = 90 \times 10,09\,\mu A = 0,91\,mA$

$V_{CE} = V_{CC} - I_C(R_C + R_E) = 22\,V - (0,91\,mA)(9,1\,k\Omega + 9,1\,k\Omega)$

$= 5,44\,V$

(b) $\beta = 135$, $I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{22\,V - 0,7\,V}{470\,k\Omega + (135)(9,1\,k\Omega + 9,1\,k\Omega)}$

$= 7,28\,\mu A$

$I_C = \beta I_B = 135 \times 7,28\,\mu A = 0,983\,mA$

$V_{CE} = V_{CC} - I_C(R_C + R_E) = 22\,V - (0,983\,mA)(9,1\,k\Omega + 9,1\,k\Omega)$

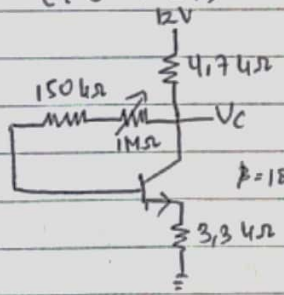
$= 4,11\,V$

(c) $\%\Delta I_C = \left|\frac{0,983\,mA - 0,91\,mA}{0,91\,mA}\right| \times 100\% = 8,02\,\%$

$\%\Delta V_{CE} = \left|\frac{4,11\,V - 5,44\,V}{5,44\,V}\right| \times 100\% = 24,45\,\%$

(d) Hasil untuk konfigurasi umpan balik kolektor lebih dekat ke konfigurasi voltage-divider daripada dua lainnya. Namun, konfigurasi voltage-divider tetap memiliki sensitif paling kecil perubahan dalam β.

25 (Fig 4.87)



1MΩ = 0Ω, $R_B = 150\,k\Omega$

$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12\,V - 0,7\,V}{150\,k\Omega + (180)(4,7\,k\Omega + 3,3\,k\Omega)}$

$= 7,11\,\mu A$

$I_C = \beta I_B = 180 \times 7,11\,\mu A = 1,28\,mA$

$V_C = V_{CC} - I_C R_C = 12\,V - (1,28\,mA)(4,7\,k\Omega) = 5,98\,V$

Semula 1MΩ, $R_B = 1000\,k\Omega + 150\,k\Omega = 1,150\,k\Omega$

$= 1,15\,M\Omega$

$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12\,V - 0,7\,V}{\Omega + (180)(4,7\,k\Omega + 3,3\,k\Omega)} = 4,36\,\mu A$

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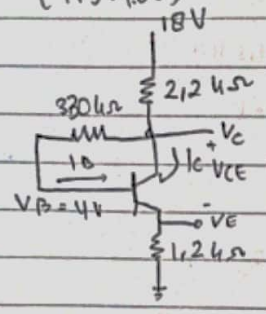
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$I_C = \beta I_B = 180 \times 4{,}136\,\mu A = 0{,}785\,mA$

$V_C = V_{CC} - I_C R_C = 12\,V - (0{,}785\,mA)(4{,}7\,k\Omega) = 8{,}31\,V$

V_C rentang dari 5,98 V ke 8,31 V

26 (Fig. 4.88)



(a) $V_E = V_B - V_{BE} = 4\,V - 0{,}7\,V = 3{,}3\,V$

(b) $I_C \cong I_E = \frac{V_E}{R_E} = \frac{3{,}3\,V}{1{,}2\,k\Omega} = 2{,}75\,mA$

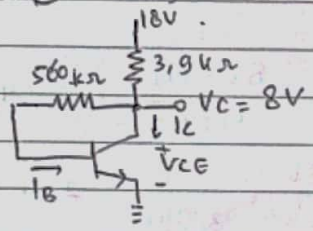
(c) $V_C = V_{CC} - I_C R_C = 18\,V - (2{,}75\,mA)(2{,}2\,k\Omega) = 11{,}95\,V$

(d) $V_{CE} = V_C - V_E = 11{,}95\,V - 3{,}3\,V = 8{,}65\,V$

(e) $I_B = \frac{V_{RB}}{R_B} = \frac{V_C - V_B}{R_B} = \frac{11{,}95\,V - 4V}{330\,k\Omega} = 24{,}09\,\mu A$

(f) $\beta = \frac{I_C}{I_B} = \frac{2{,}75\,mA}{24{,}09\,\mu A} = 114{,}16$

27 [Fig 4.89]



(a) $I_B = \frac{V_{CC} + V_{EE} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{6V + 6V - 0{,}4V}{330\,k\Omega + (121)(1{,}2\,k\Omega)} = 23{,}78\,\mu A$

$I_E = (\beta+1) I_B = 121 \times 23{,}78\,\mu A = 2{,}88\,mA$

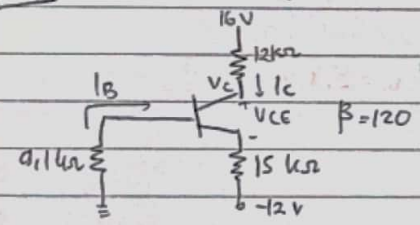
$-V_{EE} + I_E R_E - V_E = 0$

$V_E = -V_{EE} + I_E R_E$

$= -6V + 2{,}88\,mA \times 1{,}2\,k\Omega$

$= -2{,}54\,V$

28 (Fig. 4.90)



(a) $I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta+1)R_E} = \frac{12V - 0{,}7V}{9{,}1\,k\Omega + (120+1)15\,k\Omega} = 6{,}2\,\mu A$

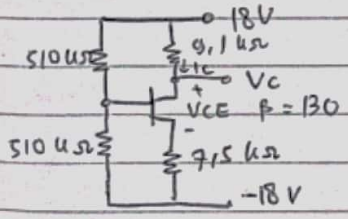
(b) $I_C = \beta I_B = 120 \times 6{,}2\,\mu A = 0{,}744\,mA$

(c) $V_{CE} = V_{CC} + V_{EE} - I_C (R_C + R_E)$

$= 16\,V + 12\,V - (0{,}744\,mA)(27\,k\Omega) = 7{,}91\,V$

(d) $V_C = V_{CC} - I_C R_C = 16\,V - (0{,}744\,mA)(12\,k\Omega) = 7{,}07\,V$

29 (Fig 4.91)



(a) $I_E = \frac{8V - 0A\,V}{2{,}2\,k\Omega} = \frac{7{,}3V}{2{,}2\,k\Omega} = 3{,}32\,mA$

(b) $V_C = 10\,V - (3{,}32\,mA)(1{,}8\,k\Omega) = 4{,}02\,V$

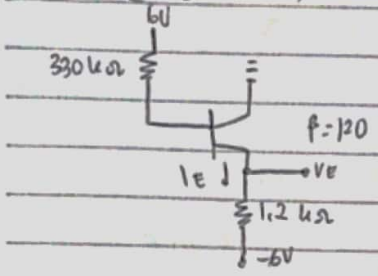
(c) $V_{CE} = 10\,V + 8\,V - (3{,}32\,mA)(2{,}2\,k\Omega + 1{,}8\,k\Omega)$

$= 4{,}72\,V$

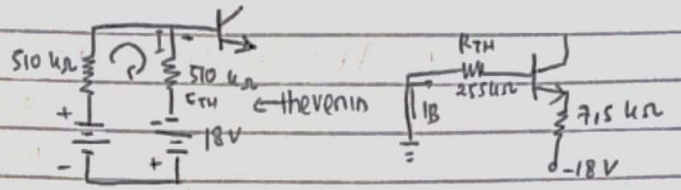
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30 (Fig 4.92)



a) $\beta R_E > 10 R_2$ not satisfied
menggunakan pendekatan yg tepat + menentukan equivalen thevenin



$R_{TH} = \frac{510\ k\Omega}{2} = 255\ k\Omega$

$I = \frac{18\ V + 18\ V}{510\ k\Omega + 510\ k\Omega} = 35,29\ \mu A$

$E_{TH} = -18\ V + (35,29\ \mu A)(510\ k\Omega) = 0\ V$

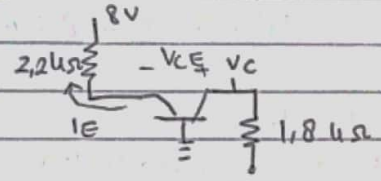
$I_B = \frac{18\ V - 0,7\ V}{255\ k\Omega + (130+1)(7,5\ k\Omega)} = 13,95\ \mu A$

b) $I_C = \beta I_B = 130 \times 13,95\ \mu A$
$= 1,81\ mA$

c) $V_E = -18V + (1,81\ mA \times 7,5 k\Omega)$
$= -4,42\ V$

d) $V_{CE} = 18V + 18V - [1,81\ mA)(0,1 k\Omega + 7,5 k\Omega)$
$= 5,95\ V$

31. (Fig 4.93)



(a) $I_B = \frac{V_{RB}}{R_B} = \frac{V_C - V_{BE}}{R_B} = \frac{8V - 0,4V}{560\ k\Omega} = 13,04\ \mu A$

(b) $I_C = \frac{V_{CC} - V_C}{R_C} = \frac{18\ V - 8V}{3,9\ k\Omega} = \frac{10\ V}{3,9 k\Omega} = 2,56\ mA$

(c) $\beta = \frac{I_C}{I_B} = \frac{2,56\ mA}{13,04\ \mu A} = 196,32$

(d) $V_{CE} = V_C = 8V$

4.8 Design Operations

32 Tentukan fixed bias jika $V_{CC} = 12\ V$, $\beta = 80$ dan $I_{CQ} = 2,5\ mA$

$I_B = \frac{I_C}{\beta} = \frac{2,5\ mA}{80} = 31,25\ \mu A$

$R_B = \frac{V_{RB}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12\ V - 0,7\ V}{31,25\ \mu A} = 361,6\ k\Omega$

$R_C = \frac{V_{RC}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{V_{CC} - V_{CEQ}}{I_{CQ}} = \frac{12\ V - 6V}{2,5\ mA} = 2,4\ k\Omega$

Nilai standard :
$R_B = 360\ k\Omega$
$R_C = 2,4\ k\Omega$

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